Topological Analysis of pLDDT Profiles: Reproducing and Extending Cazals & Sarti (2025)

Algorithms in Structural Bioinformatics — Q4 Project

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Outline

What is pLDDT and why does its *structure* matter?

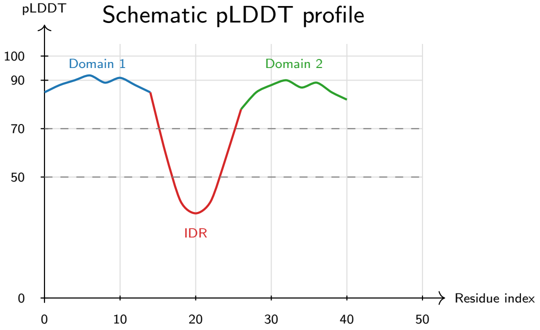
pLDDT (predicted Local Distance Difference Test)

- Per-residue confidence score, $s_i \in [0, 100]$
- Stored in the B-factor column of AlphaFold PDB files
- $s_i \gtrsim 70$: well-folded region
- $s_i \lesssim 50$: intrinsically disordered

Key observation

- Most analyses use pLDDT as a *threshold* (keep/discard)
- **Cazals & Sarti (2025)** treat the full pLDDT *profile along the chain* as a topological object
- This encodes multi-domain and disordered-linker structure

Schematic pLDDT profile



Project goal

Reproduce Q4 pLDDT fragmentation on 23 391 *H. sapiens* fragments; extend with a Q1 arity cross-correlation.

Path graph and scalar filtration

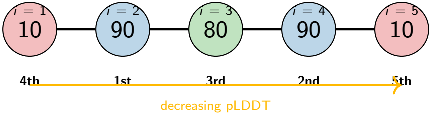
Model: protein chain $\to$ **path graph** $G_n = (V, E)$

- Vertices $V = \{1, \ldots, n\}$: residues
- Edges $E = \{(i, i{+}1)\}$: peptide bonds
- Scalar function: $u_i = -s_i$ (negative pLDDT)

Filtration: insert vertices in increasing u order (= **decreasing pLDDT** order)

$$\emptyset \subseteq \mathcal{G}_{u_1} \subseteq \cdots \subseteq \mathcal{G}_{u_n} = G_n$$

An edge $(i, i{+}1)$ is **activated** the moment *both* endpoints are inserted.



Numbers = pLDDT. Insertion order = by decreasing pLDDT.

Connected components, Elder Rule, and Persistence Diagram

$N_{cc}(u)$: number of connected components at level u

- New vertex inserted $\Rightarrow N_{cc} + 1$
- Edge joins two components $\Rightarrow N_{cc} - 1$

Elder Rule: when two components merge, the one born at *higher* pLDDT (older) survives. The younger one **dies**, producing a pair:

$$(b_i,\, d_i) \quad b_i \geq d_i \geq 0$$

Persistence: $p_i = b_i - d_i$
Accretion: $p_i = 0$ (same-level merge)

Persistence Diagram (PD)

Collection of all $m = n - 1$ pairs $\mathcal{P} = \{(b_i, d_i)\}_{i=1}^{n-1}$

Toy example: pLDDT $= [10, 90, 80, 90, 10]$

Event	N_{cc}	PD pair
Insert $i = 2$ (90)	1	—
Insert $i = 4$ (90)	2	—
Insert $i = 3$ (80), edge (2, 3)	2	$(80, 80)^{\dagger}$
edge (3, 4)	1	(90, 80)
Insert $i = 1$ (10), edge (1, 2)	1	$(10, 10)^{\dagger}$
Insert $i = 5$ (10), edge (4, 5)	1	$(10, 10)^{\dagger}$

†Accretions — zero persistence (the loser is born and dies at the same level)

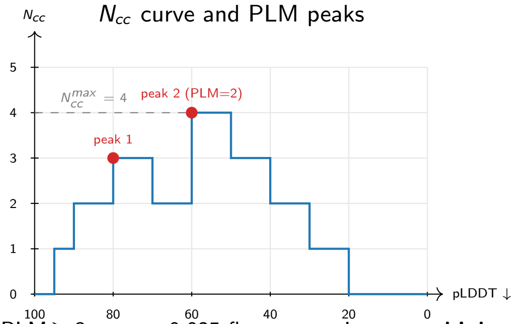
Positive pair (90, 80) at edge (3, 4): both components born at 90, tie broken by index $\Rightarrow \{4\}$ dies, pair $= (\text{birth}, \text{death}) = (90, 80)$.

$\Rightarrow f_{cp}^{+} = 1/4$, $H_p = 0.8112$ — matches the validation toy in the report.

Five topological statistics

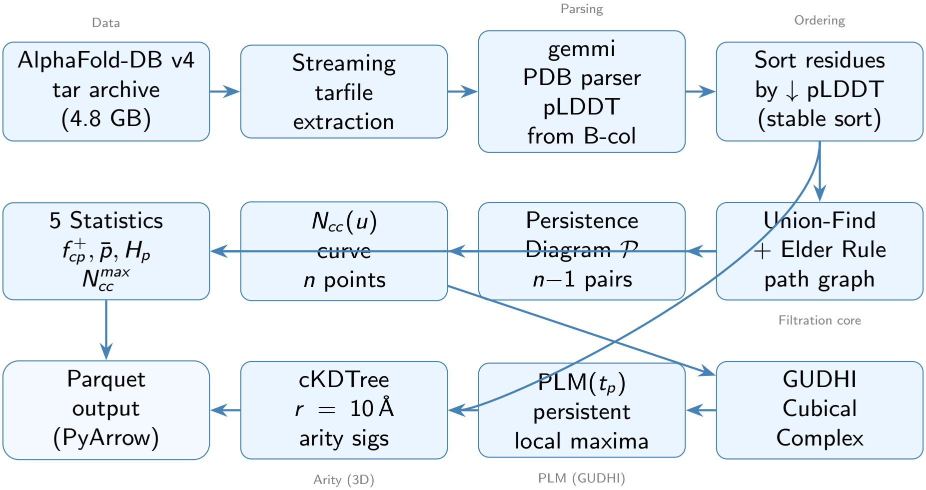
Let $m = n{-}1$, $m' =$ positive-persistence pairs, $\mathcal{P}_D =$ set of distinct persistence values.

1. $f_{cp}^{+} = \dfrac{m'}{n-1}$
   Fraction of heterogeneous transitions
2. $\bar{p} = \dfrac{1}{m} \sum_i (b_i - d_i)$
   Mean persistence lifetime
3. $H_p = \dfrac{-\sum_v \mathbb{P}[v] \ln \mathbb{P}[v]}{\ln |\mathcal{P}_D|}$
   Normalised Shannon entropy of persistence
   $H_p \in [0, 1]$; high = diverse pLDDT profile
4. $N_{cc}^{max} = \max_u N_{cc}(u)$
   Peak simultaneous fragmentation
5. $\mathrm{PLM}(t_p)$: persistent local maxima of N_{cc}
   Number of significant domain peaks



PLM ≥ 3 at $t_p = 0.025$ flags a protein as a **multi-domain fragmentation candidate**.

End-to-end pipeline overview



6 parallel workers via `multiprocessing.Pool` — processes 23 391 proteins in ≈30 seconds.

Algorithm: integer-pLDDT path-graph filtration, batched

`build_pd_and_ncc(plddt, discretise=True, batched=True)`

Input: $(s_1, \ldots, s_n)$, threshold t_p **Output:** $\mathcal{P}$, N_{cc}-curve, PLM(t_p)

1. $s_i \leftarrow \mathrm{round}(s_i)$ for all i *integer pLDDT default*
2. Init UF: $\mathrm{parent}[i] \leftarrow i$, $\mathrm{birth}[i] \leftarrow -\infty$, $\mathrm{inserted}[i] \leftarrow \perp$; $N_{cc} \leftarrow 0$
3. **for each** integer level v in decreasing $\{s_i\}$:
   - *Batch step (1):* introduce every i with $s_i = v$ as a singleton; $\mathrm{birth}[i] \leftarrow v$; $N_{cc} {+}{=} |\{i\}|$
   - *Batch step (2):* collect every incident edge $(a, a{+}1)$ with both endpoints inserted; deduplicate
   - *Batch step (3):* for each edge, in canonical order, run Elder-Rule merge
     - Same component: append (v, v) (accretion)
     - Else: loser $= \arg\min \mathrm{birth}$; $N_{cc} {-}{=} 1$; append $(\mathrm{birth}[\mathrm{loser}], v)$
   - Record $(v,\ N_{cc})$ for every i in this batch (post-batch value)
4. $\mathrm{PLM}(t_p) \leftarrow \textsc{GUDHIPersistentMaxima}(\text{Curve},\ t_\nu = n \cdot t_p)$

$\textsc{Find}$ uses path-halving. Set `discretise=False` for the raw-float ablation (changes $\mathcal{P}_D$); `batched=False` runs the sequential one-residue-per-step variant ($\mathcal{P}_D$ identical to the default; only N_{cc} and PLM differ).

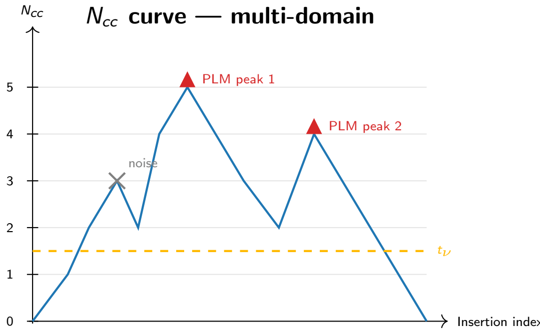
PLM via GUDHI — second-layer persistence

The N_{cc} curve is itself a 1D scalar function.
Its local maxima correspond to **pLDDT domain peaks**.

How we count significant peaks:

1. Negate N_{cc}: local maxima become local *minima*
2. Feed $-N_{cc}$ to GUDHI's `CubicalComplex`
3. Each 0-dim pair (b_j, d_j) in the resulting PD has persistence $= N_{cc}[\text{peak}] - N_{cc}[\text{valley}]$
4. Count pairs with persistence $\geq t_\nu = n \cdot t_p$

Why two layers of persistence?
First layer: filtration of the *protein graph*.
Second layer: Morse-Smale simplification of the N_{cc} *function* — turns the raw plateau structure into a sparse, threshold-stable count of persistent peaks.



Arity: a 3D packing descriptor

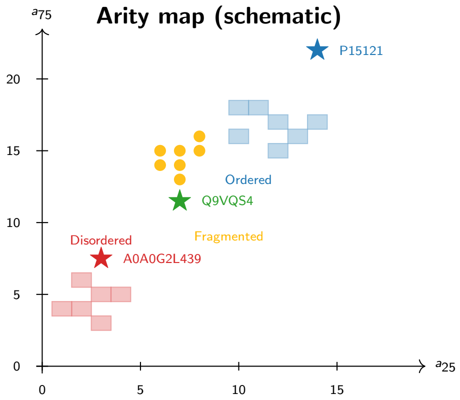
Arity of residue k**:**
$\#$ of C_α atoms within 10 Å of $\mathrm{C}_\alpha(k)$ (excluding self)

Arity signature (a_{25}, a_{75}):
25th and 75th percentiles of per-residue arity

- a_{25} low $\Rightarrow$ flexible/disordered regions dominate
- a_{75} high $\Rightarrow$ well-packed globular regions present
- Multi-domain proteins: *both* — compact domains + disordered linkers

Implementation:

- Extract C_α coordinates (gemmi)
- Build `scipy.cKDTree`
- `query_ball_point(r=10)` — $O(n \log n)$
- Subtract 1 for self-count



Validation: toy example and null model

Toy protein ($n = 5$, pLDDT $= [10, 90, 80, 90, 10]$)

Pair	(b_i, d_i)	Type
$(80, 80)$	accretion	
$(90, 80)$	$p = 10$	positive
$(10, 10)$	accretion	
$(10, 10)$	accretion	

Hand-computed:

$$H_p = 0.8112, \quad f_{cp}^{+} = 0.25$$

Code output (4 d.p.):

$$H_p = 0.8113, \quad f_{cp}^{+} = 0.25 \checkmark$$

Null random model (pLDDT $\sim \mathcal{U}[0, 100]$, 30 trials)

Paper H_p: 0.36 / 0.45 / 0.47 at $n = 100/1000/10\,000$

n	Raw float	**Integer (default)**
100	0.51	0.51
1 000	0.44 ✓	**0.46** ✓
10 000	0.41	**0.47** ✓

Only **integer pLDDT** reproduces the paper's three-point null curve (raw float matches at $n = 1000$ by coincidence, drifts at $n = 10\,000$).

$f_{cp}^{+} \to 1/3$ in both; $N_{cc}^{max}/n \approx 1/4$ confirms Conjecture 1.

Validation: three prototype proteins

UniProt	Type	n	H_p	f_{cp}^+	PLM
Paper targets (Fig. 1)					
P15121	Ordered	316	$\approx$0.04	—	—
A0A0G2L439	Disord.	449	$\approx$0.27	—	—
Q9VQS4	Mixed	781	$\approx$0.28	—	—
Default (integer, batched)					
P15121	Ordered	316	**0.147**	0.048	1
A0A0G2L439	Disord.	449	**0.433**	0.292	1
Q9VQS4	Mixed	781	**0.425**	0.297	2
Raw float ablation					
P15121	Ordered	316	0.371	0.241	1
A0A0G2L439	Disord.	449	0.466	0.339	1
Q9VQS4	Mixed	781	0.424	0.312	2

Ordering preserved (ordered $<$ mixed $\lesssim$ disordered) ✓

Q9VQS4 is *D. melanogaster*, downloaded as AlphaFold-DB v6 (v4 no longer available from EBI).

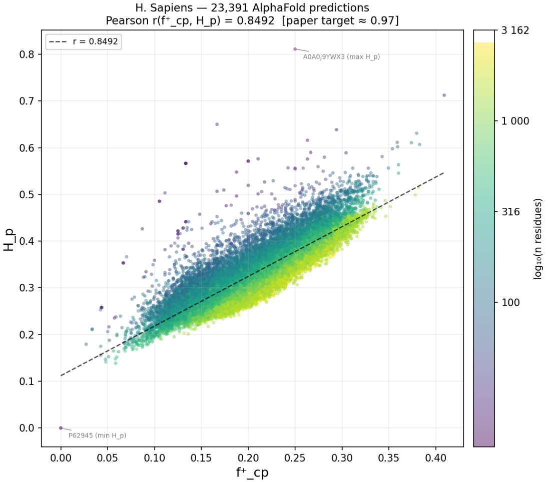
Residual ΔH_p under integer + batched

- +0.11 on P15121
- +0.16 on A0A0G2L439
- +0.15 on Q9VQS4

Integer rounding closes most of the gap on the ordered prototype (P15121 falls from 0.371 to 0.147 because its pLDDT range $[84, 99]$ collapses to 13 integer levels).

Remaining offset is attributed to **AlphaFold-DB v4 dataset drift** between the paper authors' download and the current EBI release.

Proteome-scale Pearson correlation (Figure 7)



Setup: all 23 391 AlphaFold-DB v4 fragments
No length pre-filter (paper convention)

Metric	Value
N	23 391
Pearson r (integer + batched)	**0.849**
Pearson r (raw float ablation)	0.864
Paper target	≈ 0.97
Residual gap	-0.12

r is bit-for-bit identical between sequential and batched insertion because both are **PD-derived**.

Residual gap attributed to **dataset drift** between the paper's v4 snapshot and the current EBI release.

Median $f_{cp}^{+} = 0.199$, Median $H_p = 0.317$

Fragmentation candidates and t_p sensitivity

Three-way filter (paper's Fig. 8):

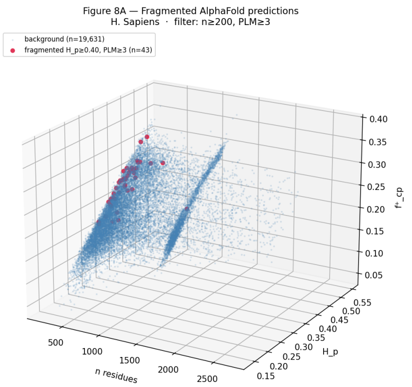
$$n \geq 200, \quad H_p \geq 0.25, \quad \mathrm{PLM}(t_p) \geq 3$$

t_p	Batched (default)	Sequential	Paper
0.020	207	759	—
0.025	**43**	164	$\approx$**86**
0.030	4	34	—

Paper's 86 sits between our two variants
$\sqrt{43 \times 164} \approx 84$.

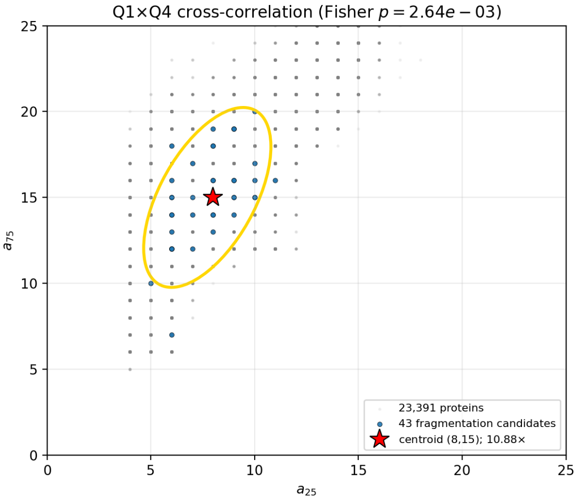
Batched insertion suppresses intra-batch N_{cc} transients $\rightarrow$ PLM tail collapses to 0 for PLM≥ 5 (was 30 under sequential).

Raw-float ablation (`discretise=False`) gives 183 candidates at $t_p = 0.025$. Residual gap to 86 attributed to dataset drift (Section: Discussion).



3D scatter: length $\times$ H_p $\times$ f^+_{cp}. Orange = fragmentation candidates.

Q1 × Q4 cross-correlation



Finding: fragmented proteins (PLM ≥ 3) cluster in a specific arity bin

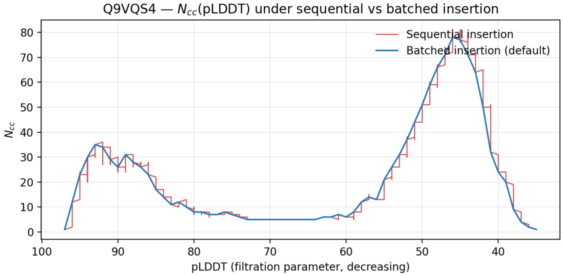
Property	Value
Candidates	43 fragmented
Centroid	$(a_{25} = 8,\ a_{75} = 15)$
Enrichment	**10.88×**
p-value	2.64×10^{-3}
Test	Fisher exact

Biological interpretation

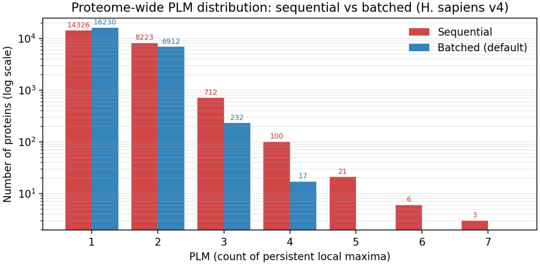
- $a_{25} \approx 8$: disordered linkers with few contacts
- $a_{75} \approx 15$: well-packed domain cores
- Intermediate arity = mix of compact domains + disordered linkers = **multi-domain hallmark**

This finding is *orthogonal* to pLDDT — derived purely from 3D C_α coordinates.

Sequential vs batched: identical PD, different N_{cc} and PLM



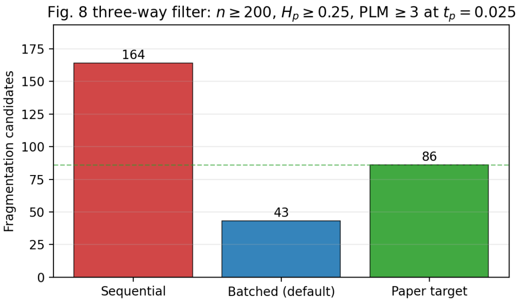
$N_{cc}(\mathrm{pLDDT})$ on Q9VQS4: sequential records intra-batch transients; batched gives flat plateaus.



PLM histogram across 23 391 proteins. Sequential PLM ≥ 5 tail: 30 proteins. Batched: 0.

Under the Elder Rule on a path graph, both variants produce *identical persistence diagrams*; only the N_{cc} curve and PLM differ. The paper says “pLDDT values come in batches” — we follow that as the default.

Fig. 8 candidate count: the paper sits between our variants



Variant	Candidates
Batched (default)	**43**
Sequential ablation	164
Paper reported	≈ 86
$\sqrt{43 \times 164}$	≈ 84

Two legitimate edge-tie conventions on the path graph; neither is “wrong”.

The paper’s count sits within rounding of the geometric mean of our two variants — strong evidence that the residual gap is **dataset drift**, not an algorithmic bug.

AlphaFold-DB v6 replication: gap is not version-specific

Question: does v6 close the gap with the paper?

Full pipeline re-run on v6 tarball (5.1 GB, May 2026)
23,586 proteins processed, 0 failures
(integer + batched defaults on both)

Metric	v4	v6	Δ
N proteins	23,391	23,586	+195
Pearson r	0.849	0.850	+0.001
Median f_{cp}^{+}	0.199	0.199	± 0
Median H_p	0.317	0.317	± 0
Candidates†	43	43	± 0

† $n \geq 200$, $H_p \geq 0.25$, PLM$(0.025) \geq 3$

Answer: no — and that confirms drift.

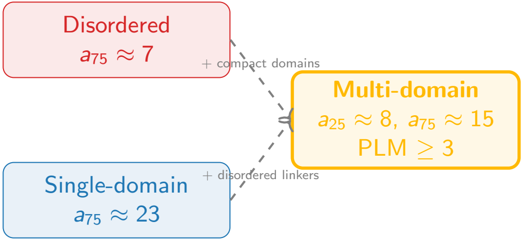
Every reported statistic is essentially **identical** between v4 and v6:

- $\Delta r = +0.001$
- Δ candidates $= 0$
- Medians coincide to 3 decimals

The residual gap to the paper is **not a release-version artifact**. It is consistent with drift between the 2024 v4 snapshot used by the paper authors and the rebuilt v4 tarball currently distributed by EBI.

Arity enrichment: biological significance

What the enrichment tells us



Two orthogonal signals agree:

1. **pLDDT fragmentation** (Q4):
   PLM ≥ 3 identifies proteins with multiple confident domains separated by linkers
2. C_α **arity** (Q1):
   Moderate a_{25} (linkers) + moderate a_{75} (domains) $\rightarrow$ same proteins

Convergence of two independent structural descriptors provides mutual validation and strengthens confidence in the topological fragmentation signal.

Summary

What we reproduced

- Path-graph filtration from scratch (Union-Find + Elder Rule, integer pLDDT, batched insertion)
- All 5 statistics + PLM via GUDHI
- Applied to all 23 391 H. sapiens AlphaFold-DB v4 fragments
- Pearson $r = 0.849$ vs target 0.97
- 43 Fig. 8 candidates vs target 86 (sequential ablation: 164; $\sqrt{43 \times 164} \approx 84$)
- Null model: paper $H_p \approx 0.36$ / 0.45 / 0.47 reproduced at $n = 100/1000/10\,000$ ✓

Novel extension (Q1 × Q4)

- Proteome-wide arity signatures
- 10.88-fold enrichment of fragmented proteins at $(a_{25} = 8, a_{75} = 15)$
- $p = 2.64 \times 10^{-3}$ (Fisher exact)
- Orthogonal 3D validation of pLDDT fragmentation signal

Key insight

Integer pLDDT + batched insertion reproduce the paper's algorithm; the residual gap to per-protein numbers is dataset drift, not an algorithmic bug.

Code: Python 3.11 — NumPy · SciPy · gemmi · GUDHI · PyArrow · streaming tar pipeline (30 s for 23 391 proteins)

Thank you

Questions?

Paper: Cazals & Sarti (2025), bioRxiv 2024.11.16.623929
Data: AlphaFold-DB v4 — *H. sapiens* proteome